



Elettra Sincrotrone Trieste

TRACKING-BASED LINEAR SYMPLECTIC PARAMETERIZATION OF INSERTION DEVICES

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Abstract

Accurate linear models of insertion devices (IDs) are essential for optics correction, coupling control, and feed-forward strategies in modern light sources. We present a systematic framework that builds a linear transverse symplectic parameterization, including leading-order chromatic effects, directly from Radia's Runge–Kutta (RK) particle tracking. We compare the resulting matrices with those obtained using the standard kick-maps approach for planar and elliptic IDs. The resulting parameterized symplectic models are well suited for constructing feed-forward tables across ID operating ranges.

Radia

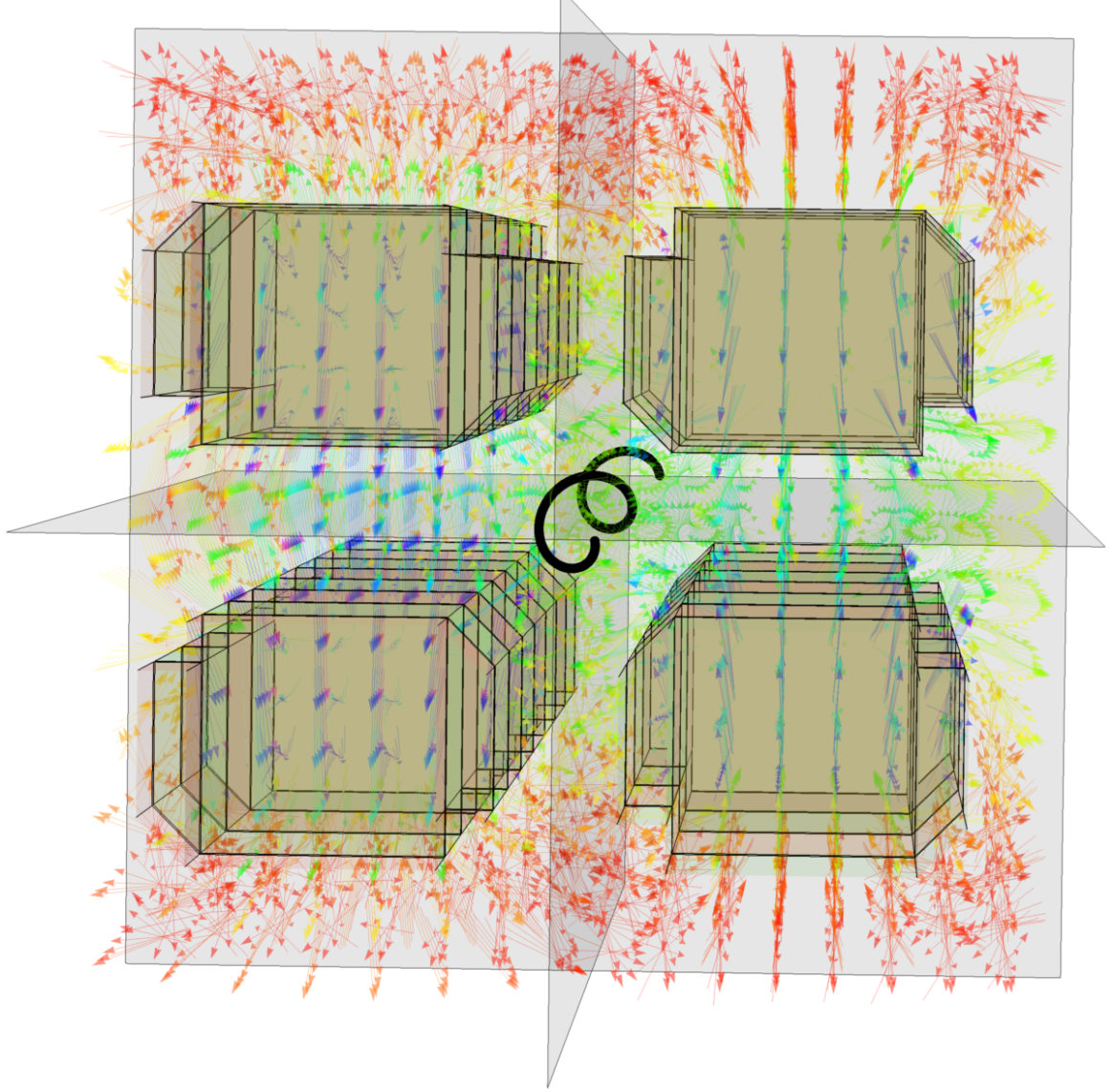


Figure 1: 3D Radia model (one ID period) with magnetic field and reference trajectory.

Elleau's potential and transverse kicks

Magnetic field (Radia):

$$\mathbf{B}(x, y, s) = (B_x(x, y, s), B_y(x, y, s), B_z(x, y, s))$$

Elleau's potential:

$$I_x(x, y, s) = \int_{-\infty}^s B_x(x, y, s') ds'$$

$$I_y(x, y, s) = \int_{-\infty}^s B_y(x, y, s') ds'$$

$$\Phi(x, y, s) = I_x^2(x, y, s) + I_y^2(x, y, s)$$

$$U(x, y) = \oint \Phi(x, y, s) ds$$

Angular kicks ($\alpha = \frac{e}{\gamma mc}$):

$$K: \quad \Delta x' = -\frac{\alpha^2}{2} \frac{\partial U(x, y)}{\partial x}, \quad \Delta y' = -\frac{\alpha^2}{2} \frac{\partial U(x, y)}{\partial y}$$

Trajectories

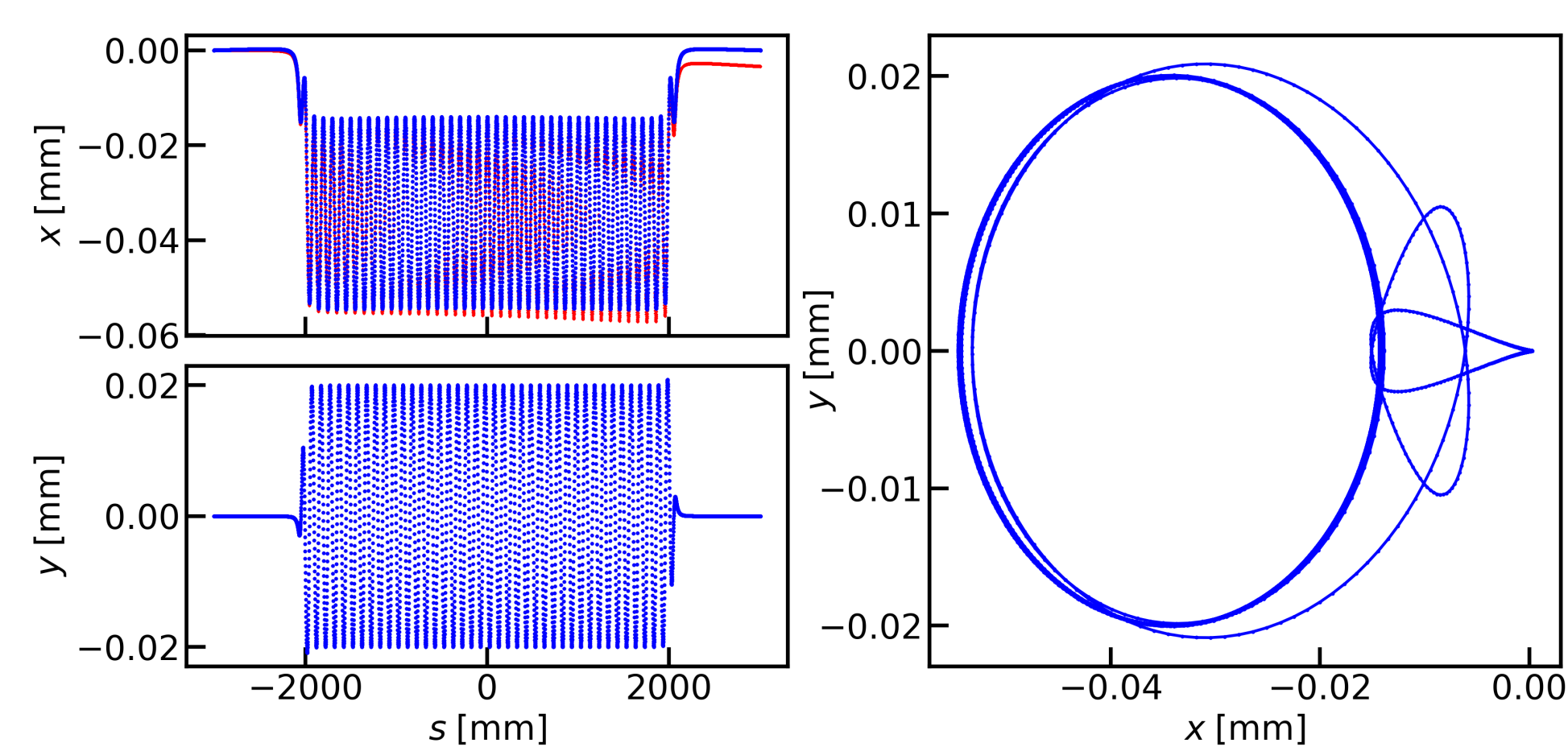


Figure 2: Particle trajectory in the EU100 (APPLE-II) insertion device for unmatched (red) and matched (blue) orbits.

Radia's RK integrator & canonical tracking

Equations of motion (Radia):

$$x'' = -\frac{\alpha}{\sqrt{1+x'^2+z'^2}} [z'B_y + (1+x'^2)B_z + x'z'B_x]$$

$$z'' = +\frac{\alpha}{\sqrt{1+x'^2+z'^2}} [x'B_y + (1+z'^2)B_x + x'z'B_z]$$

Canonical coordinates $(x, x', y, y') \iff (q_x, p_x, q_y, p_y)$:

$$x' = \frac{p_x}{\sqrt{(1+\delta)^2 - p_x^2 - p_y^2}}, \quad y' = \frac{p_y}{\sqrt{(1+\delta)^2 - p_x^2 - p_y^2}}$$

Canonical integration M_{ID} :

$$(q_x, p_x, q_y, p_y) \Rightarrow (x, x', y, y')$$

$$(x, x', y, y') \Rightarrow \text{RK} \Rightarrow (x, x', y, y')$$

$$(x, x', y, y') \Rightarrow (q_x, p_x, q_y, p_y)$$

Thin-lens insertion:

$$\hat{M}_{ID} = D_{L/2}^{-1} \circ M_{ID} \circ D_{L/2}^{-1}$$

Potential model:

$$M_{ID} = \prod_{i=1}^{n_p} [D_{\ell_p/2} \circ K \circ D_{\ell_p/2}]$$

Linear ID model parameterization

Linear matrix (finite difference $\Lambda(\varepsilon) = \varepsilon I$):

$$\hat{M} = \frac{1}{2\varepsilon} [\varphi(\Lambda(+\varepsilon)) - \varphi(\Lambda(-\varepsilon))]^T$$

Exponential representation:

$$M = \exp(SA), \quad A = A^T \iff H = \frac{1}{2} \mathbf{x}^T A \mathbf{x}, \quad \mathbf{x} = (q_x, p_x, q_y, p_y)^T$$

$$\hat{A} = -S \log(\hat{M})$$

Leading order chromatic linear matrix:

$$M = \exp(SA) \exp(\delta SB)$$

$$\hat{B} = -S \exp(-\hat{S}\hat{A}) \left[\frac{M(\delta=+\Delta) - M(\delta=-\Delta)}{2\Delta} \right]$$

$$\hat{B}_{sm} = \frac{1}{2} (\hat{B} + \hat{B}^T)$$

$$M_{ID}(\delta) = \exp(\hat{S}\hat{A}) \exp(\delta \hat{S}\hat{B}_{sm})$$

Skew diagonal elements (coupling)

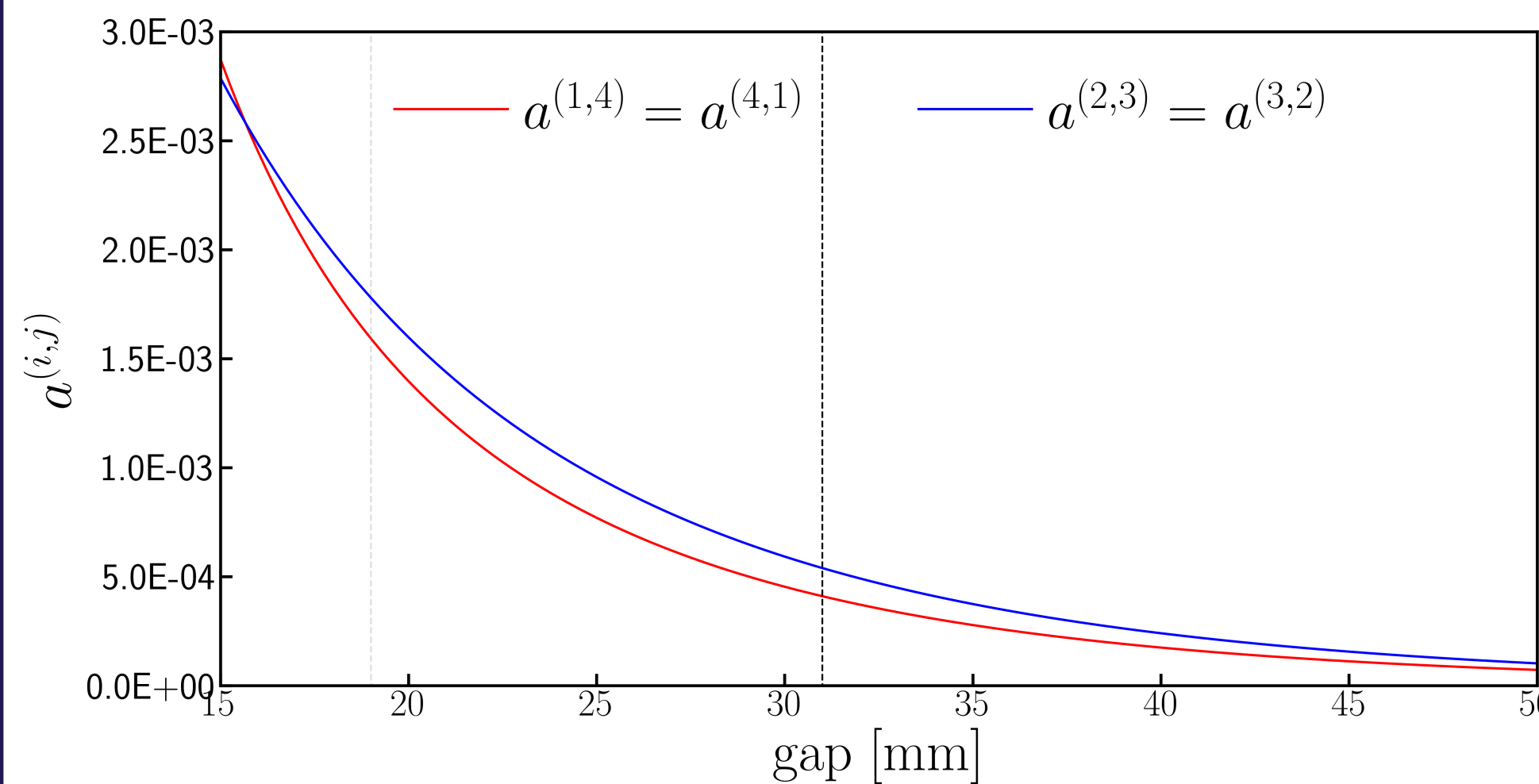


Figure 4: Skew diagonal elements vs ID gap size.

Linear optics distortion

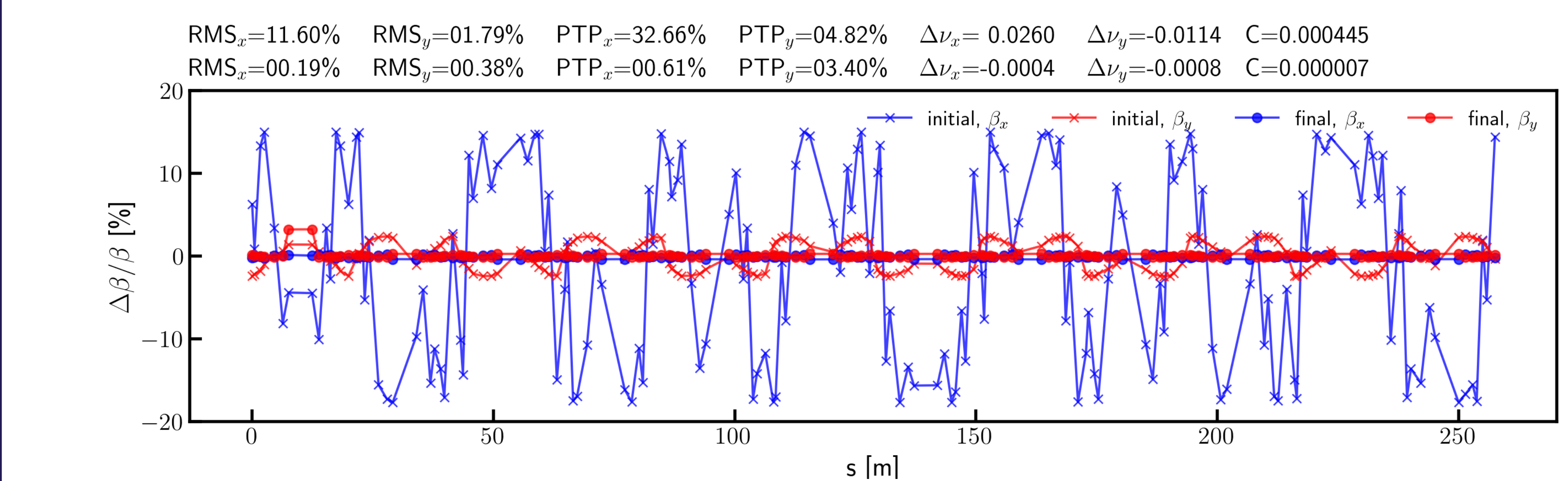


Figure 6: Beta-beating generated by EU100 before and after correction.

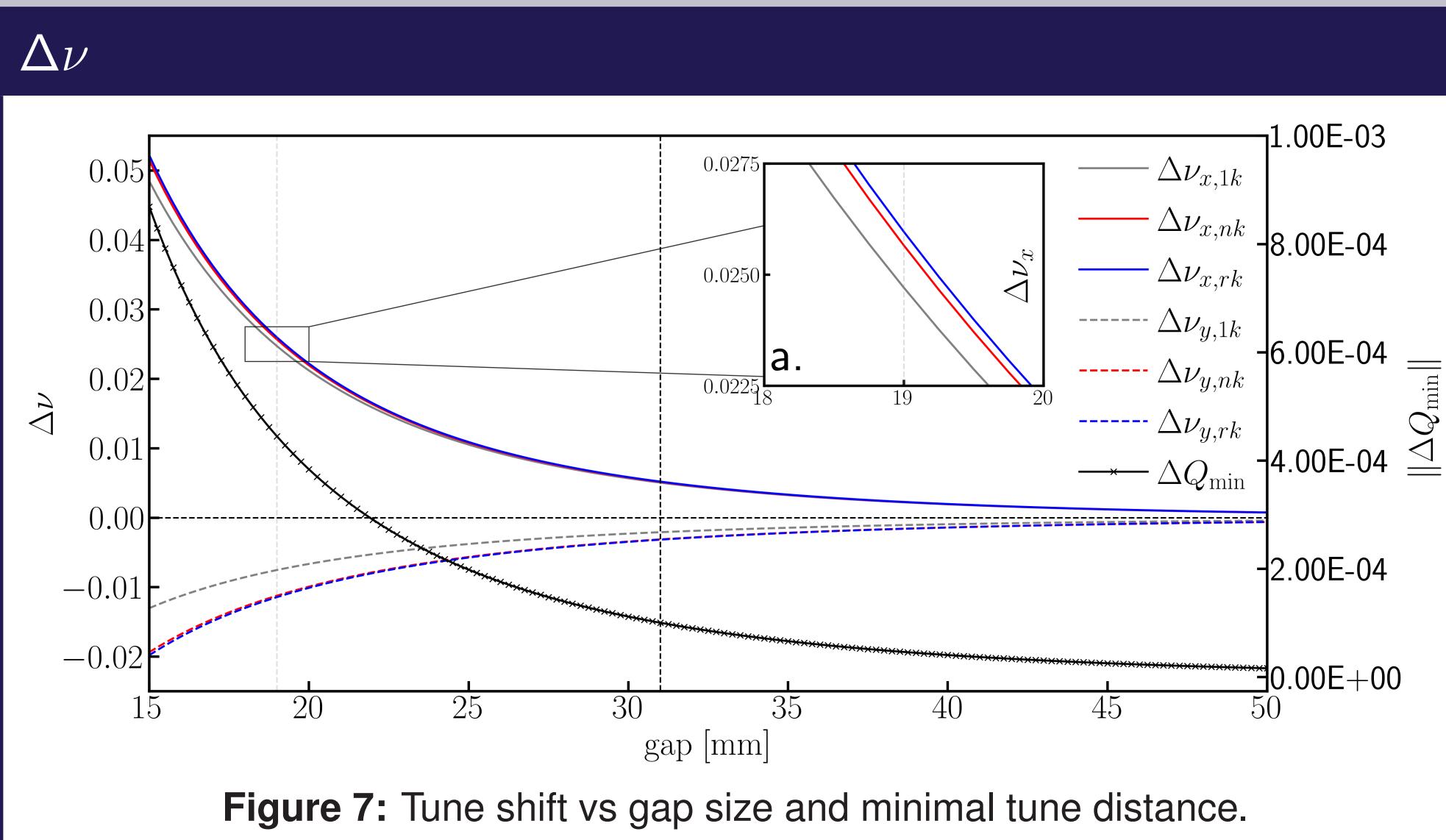


Figure 7: Tune shift vs gap size and minimal tune distance.

Optics distortion (gap & phase scan)

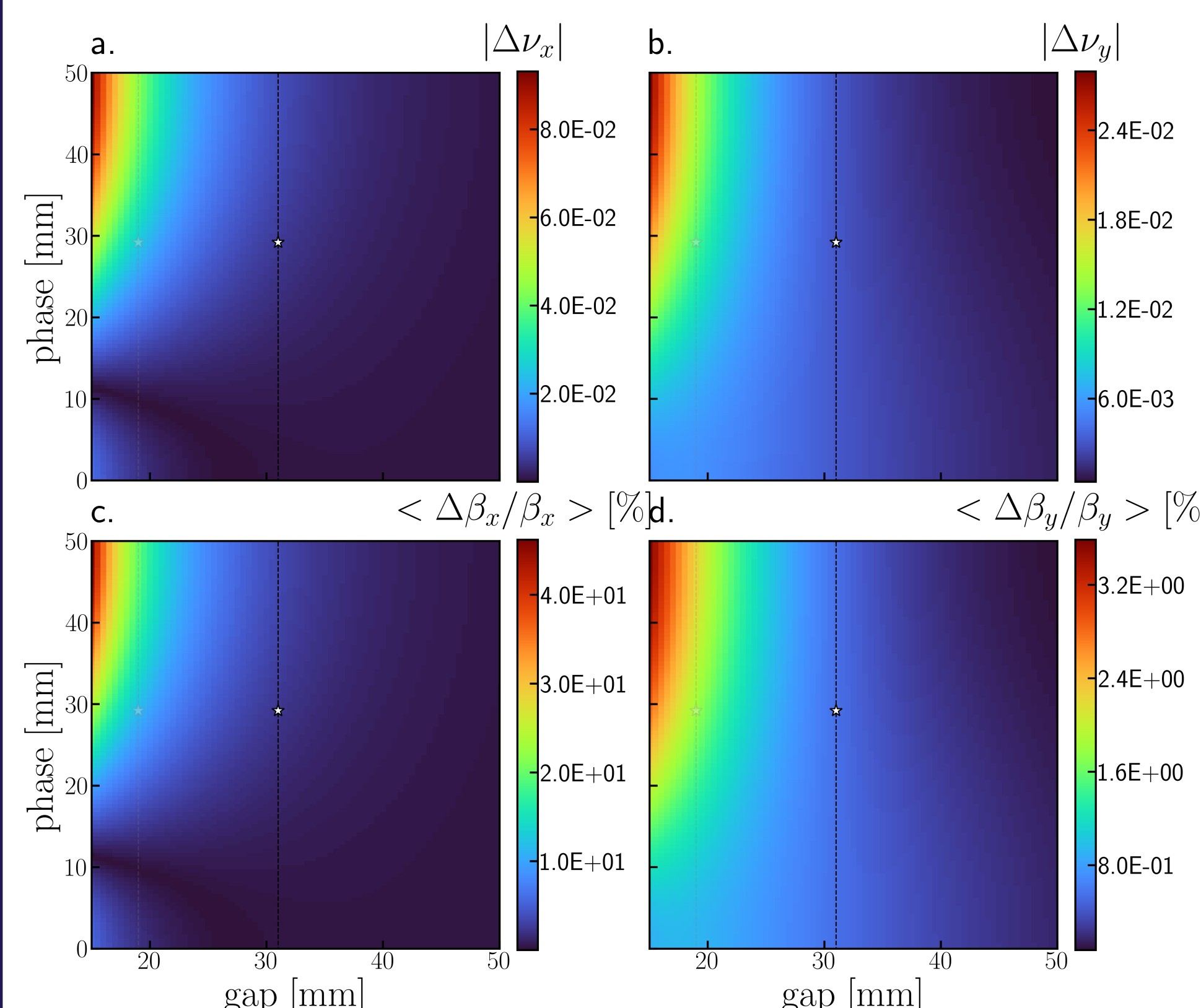


Figure 9: Tune shift and RMS beta-beating vs ID parameters.

Main diagonal elements (focusing)

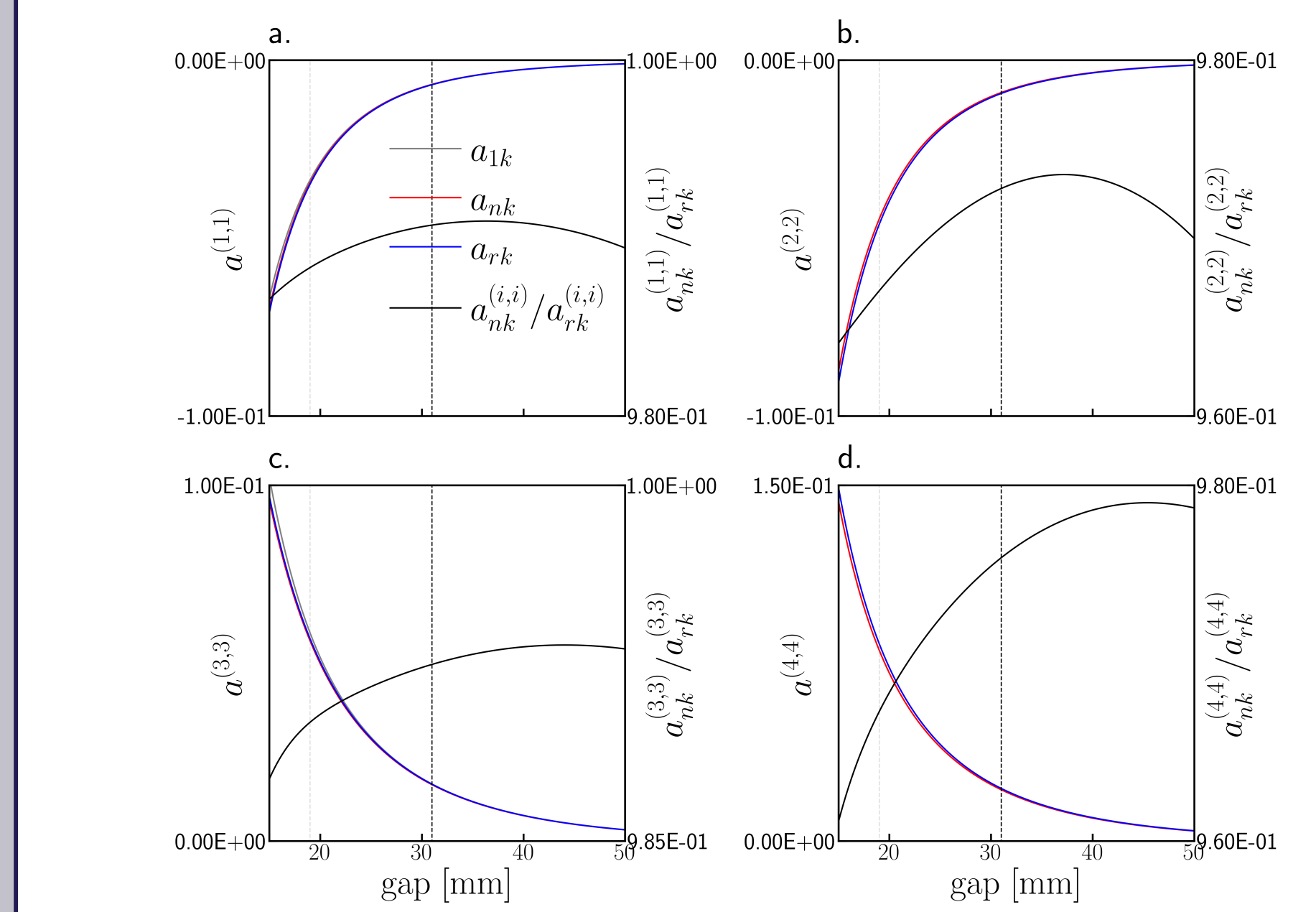


Figure 3: Main diagonal elements vs ID gap size comparison.

Main diagonal elements (gap & phase scan)

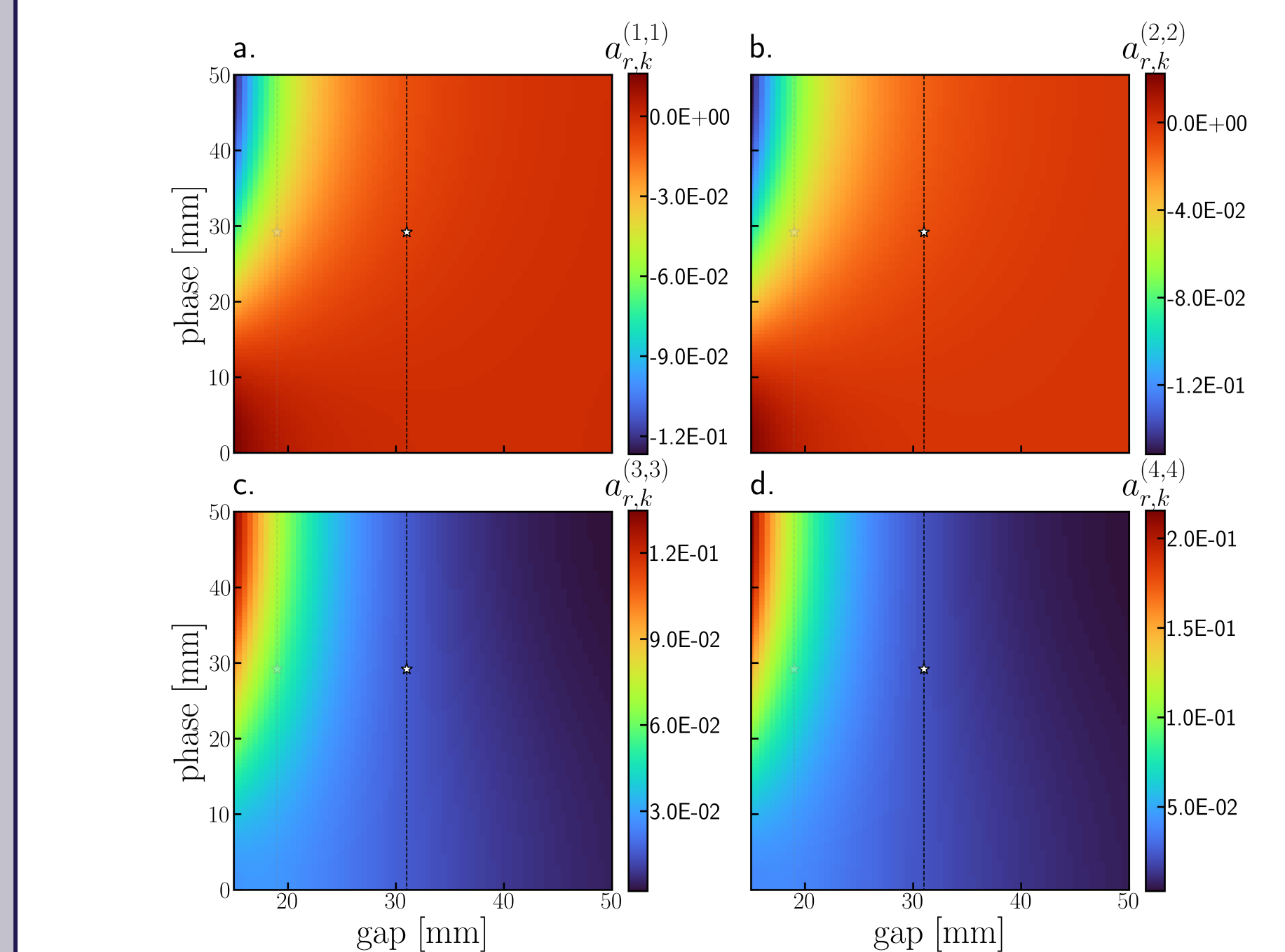


Figure 5: Main diagonal elements vs ID gap and phase scan.

Delta beta / beta

EU100 (APPLE-II)

ℓ_p 100 mm
 n_p 40
gap 31 mm
(k_x, k_y) (3.9, 6.6)

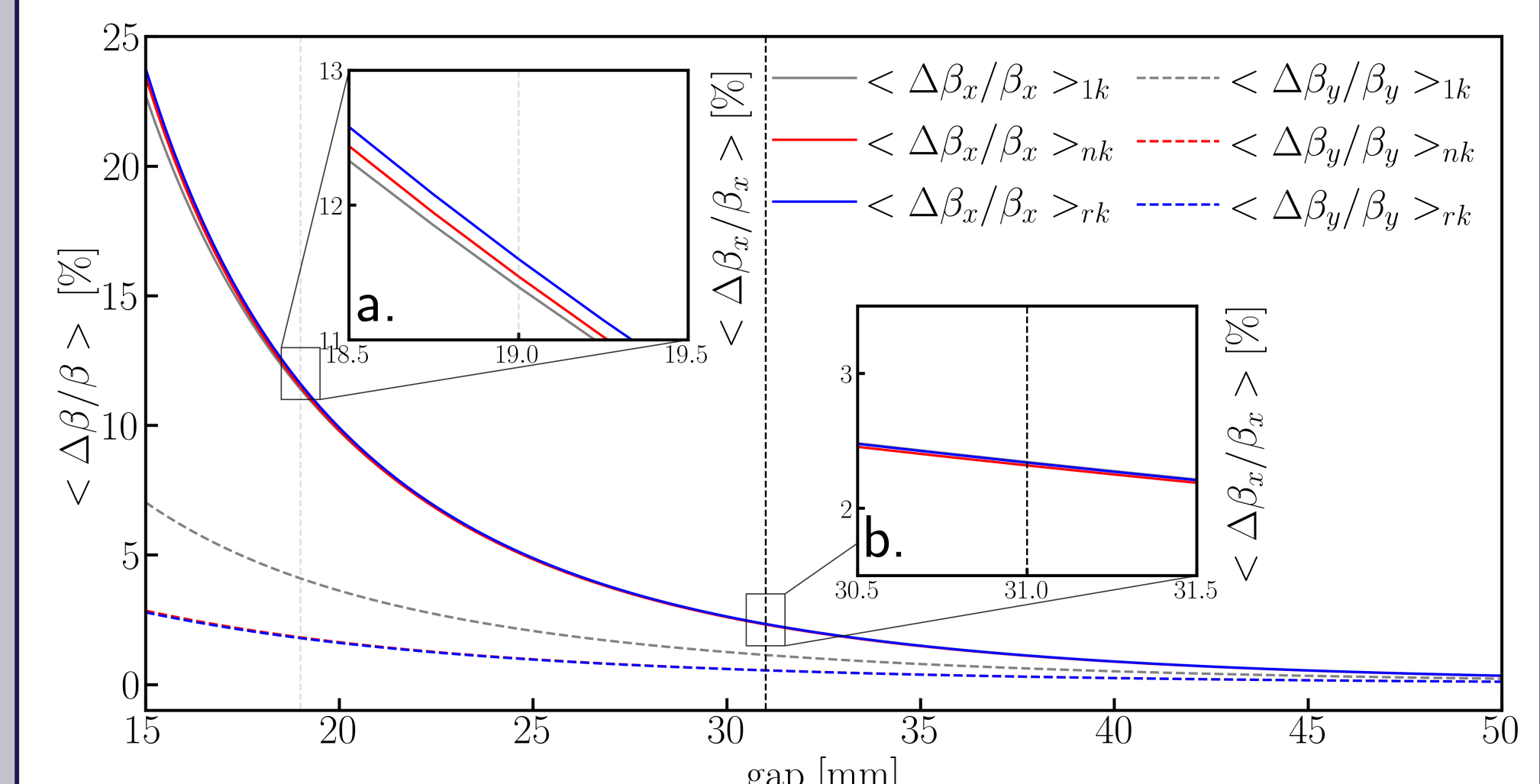


Figure 8: Beta-beating vs gap size.

Conclusion

Linear ID models based on Elleau's potential (a single kick and one kick per period) and on the RK integrator were investigated for different Elettra 2.0 IDs. Differences in the resulting optics distortion between the NK and RK models were found to be small and within Elettra 2.0 tolerances (RMS beta-beating < 5%). The RK model also includes linear coupling, which is small but may become relevant for future light sources.

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MATRIX